Search for time-reversal-symmetry violation in thallium fluoride using a cryogenic buffer-gas beam source

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Abstract

The CENTREX experiment aims to provide a 30-fold increase in sensitivity over the best present upper bounds on the strength of time-violating fundamental interactions by measuring shifts in nuclear resonant frequencies in thallium fluoride. I describe the motivation for studying these time-violating interactions, and provide an overview of past work. I present the current measurement scheme for this experiment. Since the apparatus is still under construction, I present the components already built, give preliminary data collected to date, and estimate the timeline for completion of the experiment.

Contents

1 Rotational Cooling **1**

1.1 Required microwave power 1

1.2 Vacuum chamber 2

1.2.1 AR windows 2

1.2.2 Clipping 8

1.2.3 Beam-blocking mesh 9

1.3 Microwave system 10

References **11**

List of Figures

1 Window reflectivity vs microwave frequency. 4
2 Residual microwave reflectivity when the window ($t = 11.494$ mm) is not of the optimal thickness. 5
3 Residual microwave reflectivity versus refractive index variations, for window thickness $t = 11.494$ mm. 6
4 Residual reflectivity versus microwave frequency variations, for window thickness $t = 11.494$ mm. 6
5 Residual reflectivity versus angle of incidence. 7
6 Residual reflectivity due to wavefront curvature. 8
7 RC chamber 9
8 Transmission of 13.4 GHz microwaves at normal incidence through a wire mesh. 10

List of Tables

1 Rotational Cooling

1.1 Required microwave power

In this section, we work out an estimate of the microwave power needed to drive rotational cooling transitions.

Matrix elements In a simplified model of a diatomic polar molecule [1, sect 7.6], consisting of a rigid rotor, the eigenstates are states with definite values of the angular momentum J and its z-projection, m. In the center of mass frame of the rotor, the wavefunction is the spherical harmonic $Y_J^m(\theta, \phi)$.

The molecule-fixed dipole moment of TlF, i.e., the dipole moment along the symmetry axis of the molecule, is 4.2 Debye = 1.7 ea_0 [2, Table 1]. Immediately making the rotating-wave approximation, the Rabi frequency due to the applied microwave electric field E_0 Hamiltonian's interacting with the dipole is (cf. [1, eqn 7.81])

$$\Omega(J', m', J, m) = \frac{dE_0}{2\hbar} \int_0^{2\pi} \mathrm{d}\phi \int_0^{\pi} \mathrm{d}\theta \; Y_{J'}^{*m'}(\theta, \phi) \cdot \cos\theta \cdot Y_J^m(\theta, \phi) \cdot \sin\theta$$

The integral will in general be either zero or some order-unity constant. For example, for the $J = 3 \to 2$ transitions the integrals are

$$\frac{\Omega(2, m', 3, m)}{dE_0/2\hbar} \approx \begin{pmatrix} 0 & 0.378 & 0 & 0 & 0 & 0 \\ 0 & 0.478 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.507 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.478 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.378 & 0 \end{pmatrix}_{m'm}$$

We see that only the $\Delta m = 0$ elements are nonzero. Finding all such elements for $J \leftrightarrow J+1$

for $J = 0, 1, 2, 3$, we get:

$$\frac{\Omega(J+1, m, J, m)}{dE_0/2\hbar} \approx \begin{pmatrix} 0 & 0 & 0 & 0.578 & 0 & 0 & 0 \\ 0 & 0 & 0.447 & 0.516 & 0.447 & 0 & 0 \\ 0 & 0.377 & 0.478 & 0.507 & 0.478 & 0.377 & 0 \\ 0.333 & 0.436 & 0.487 & 0.503 & 0.487 & 0.436 & 0.333 \end{pmatrix}_{Jm}$$

Thus, the Rabi frequency is approximately

$$\Omega \approx 0.4 \frac{dE_0}{2\hbar}.$$

Fly-through time, or excited state lifetime Assuming (as an order of magnitude estimate) the rotational cooling region is about 10 cm long, the flythrough time is

$$t_0 = \frac{0.02 \text{ m}}{200 \text{ m/s}} = 0.1 \text{ ms}.$$

Microwave power Rearranging $\Omega \gg 1/t_0$, the microwave electric field must be much greater than

$$E_0 = \left(\frac{2\pi}{t_0}\right) / \left(\frac{0.4\ d}{2\hbar}\right) = 2.3 \text{ V/m}.$$

This corresponds to a minimum intensity

$$I_{\min} = \frac{c\epsilon_0}{2} E_0^2 = 7.0 \text{ mW/m}^2.$$

Given a waist size $w \approx 5$ cm, the microwave power required will have to be much greater than $P_{\min} = \pi w_0^2 I_{\min} = 0.05$ mW.

1.2 Vacuum chamber

1.2.1 AR windows

The windows of the vacuum chamber were designed to reflect as little of the microwaves as possible while still being strong enough to withstand the atmospheric pressure.

Pressure window thickness The thickness of a circular window of radius R required to withstand a certain pressure p is given by [3, eqn 4.190]

$$t = R\sqrt{\frac{Kp\,\mathrm{Sf}}{Y}},$$

where Sf is the safety factor (4 is adequate unless the deformation of the window would lead to unacceptable waveform distortion). The modulus of rupture Y is 7,000 psi for fused silica, 65,000 psi for sapphire. K is a constant whose value can be taken as 0.75 for a clamped edge and 1.125 for a free edge. Thus, for example, to withstand atmospheric pressure (14.7 psi), a clamped fused silica window of radius 5 cm would have to have a thickness of at least 8 mm.

Dielectric loss The fraction of power transmitted through a window is subject to dielectric loss in the window. As it propagates a distance z into the window, the power (normalized to input power) decays as

$$P(\delta, \lambda, z) = \exp\left(-\delta\frac{2\pi}{\lambda}z\right),$$

where the wavelength in the dielectric material is $\lambda = c/(nf)$, and δ is the loss angle, commonly quoted as the loss tangent $\tan\delta$. For fused silica, $\tan\delta \approx 0.0003$. The dielectric loss, which causes the frequency dependence, is a negligible effect. E.g., going through a 10 mm thick window, it would be about 0.2% to 0.6% at our frequencies. Since this is merely radiation lost as heat (as opposed to microwaves), a 1% power loss is tolerable and can be ignored in what follows.

Coherence length For light passing through a glass window, we normally cannot observe thin film interference, since the coherence length or sunlight (≈ 0.6 μm) is much smaller than the window thickness (a few millimeters).

However, coherence length being inversely proportional to the bandwidth, it is much greater in the microwave range. For example, a source with a 1 MHz frequency uncertainty would have a vacuum coherence length of $L = c/1\ \mathrm{MHz} = 300$ m.

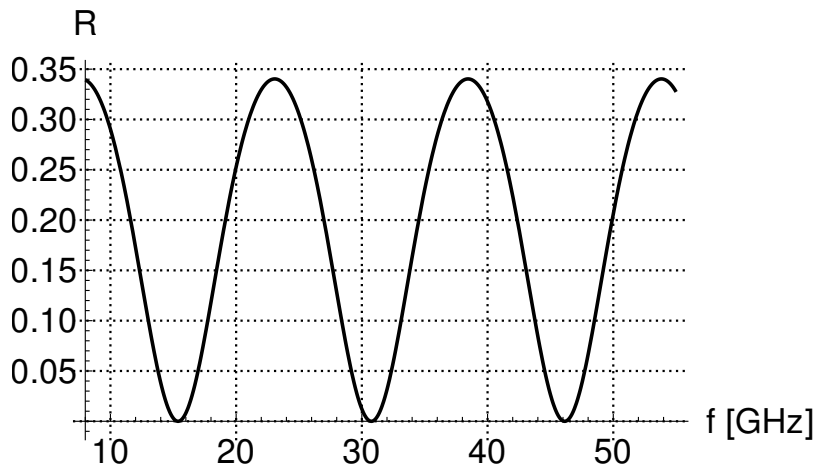

Figure 1: Reflectivity of a 5 mm thick fused silica ($\epsilon_\mathrm{r} = 3.8$) window exposed to microwave radiation at normal incidence.

Thin film reflectance at normal incidence The phase change for planar wavefronts at normal incidence across a window of thickness d and refractive index n will be $\delta = 2\pi nd/\lambda_0$, where λ_0 is the vacuum wavelength. The admittance of the window is given by [4, eqn 2.93]

$$Y = \frac{y_0 \cos\delta + i y_1 \sin\delta}{\cos\delta + i(y_0/y_1)\sin\delta}, \tag{1}$$

where y_0 is the admittance of free space, while for an ideal dielectric window, its admittance will be $y_1 = \sqrt{\epsilon_1/\mu_1}$. Since for dielectrics, $\mu \approx \mu_0$, we can assume $y_1 \approx \sqrt{\epsilon_\mathrm{r}} y_0$. For fused silica, $y_1 \approx \sqrt{3.8} y_0$.

The corresponding power reflectivity is [4, eqn 2.91]

$$R = \left|\frac{y_0 - Y}{y_0 + Y}\right|^2 \tag{2}$$

A 5 mm thick fused silica window would thus have the reflectivity shown in Fig. 1. We see that unless the window thickness is matched to the frequencies of the radiation we use, the window can reflect up to 35% of incident power. However, when the frequency is matched, ideally no power is reflected at all. Even if we change the frequency by 1.6 GHz from each minimum, the reflectivity will stay below 5%.

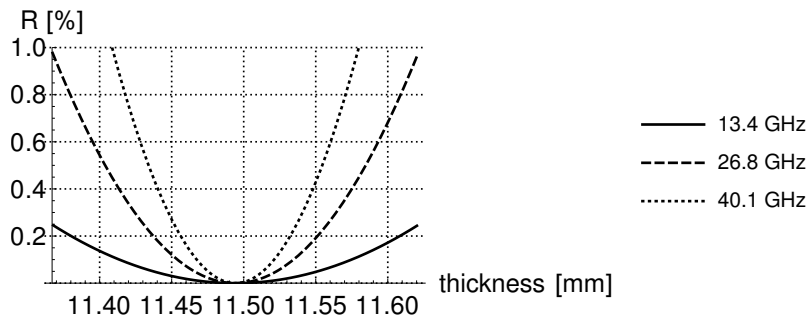

Figure 2: Residual microwave reflectivity when the window ($t = 11.494$ mm) is not of the optimal thickness.

Half-wave windows Due to the periodicity of the reflection coefficient, we only need a single window thicknesses to cover our three microwave frequencies (13.4 GHz, 26.8 GHz, 40.1 GHz), namely the half-wave thickness 5.75 mm (or a multiple of this thickness). Indeed, it is a well known fact that a stack containing an integer number of half-wave layers has no effect on reflectance or transmittance of assemblies, at wavelengths at which they are truly half-wave. Thus "half-wave layers are sometimes referred to as absentee layers" [4, sect 2.9].

As explained above, the window has to be at least 8 mm thick to withstand the atmospheric and other forces. That is why an integer multiple of the thickness has to be chosen, such as 2×5.747 mm $= 11.494$ mm. Unfortunately, the windows used in CENTREX to date (Jun 2023) were ordered as (11.762 ± 0.12) mm thick owing to a confusion in calculating the wavelength in vacuum vs in the material ($\sqrt{3.8} \approx 1.95 \approx 2$). Besides, it is not clear how close to 3.8 the relative permittivity actually is; it would be interesting to measure the reflectivity of several windows in the thickness range of, say, 5% around the predicted optimal thickness of 11.50 mm.

Residual reflectivity The lower bound for reflectivity is realistically set by how precisely the window thickness can be controlled. It can be kept below 1% if the window thickness can be within about ± 2 mil $= 0.05$ mm, as seen in Fig. 2.

Of course, this assumes a particular value of the refractive index. For fused silica, the refractive index is is 1.812 ± 0.003 at 94.3 GHz, but 1.949 at 105.3 GHz. [5, Table 5.1] Whatever the material used, its refractive index needs to be known precisely in order to determine the exact optical thickness of the window (Fig. 3).

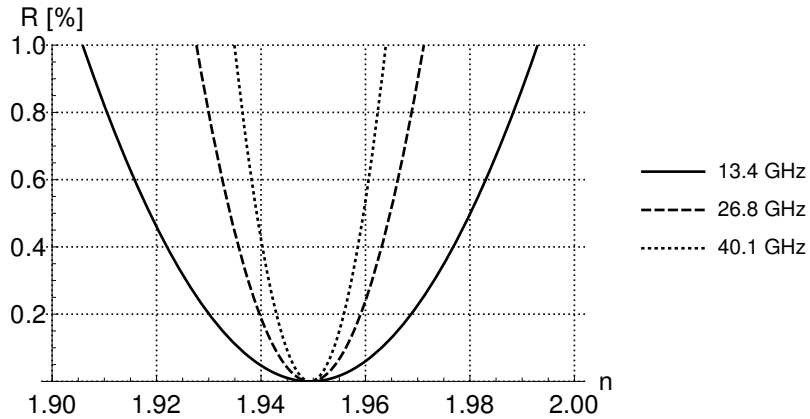

Figure 3: Residual microwave reflectivity versus refractive index variations, for window thickness $t = 11.494$ mm.

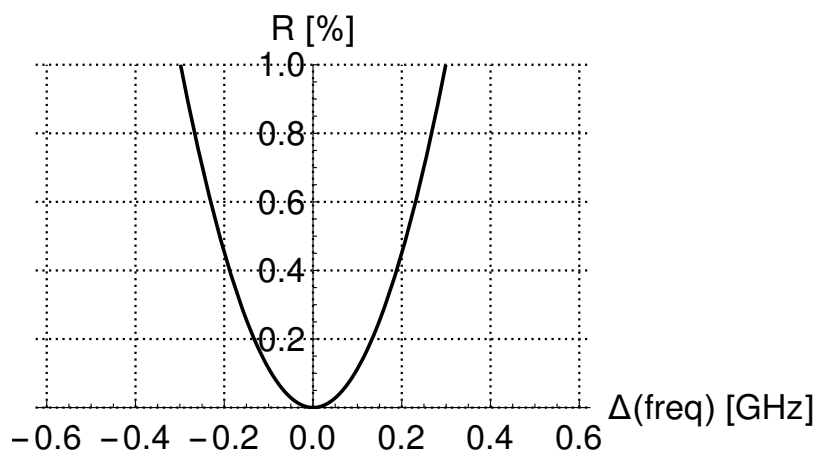

Figure 4: Residual reflectivity versus microwave frequency variations, for window thickness $t = 11.494$ mm.

The frequency bandwidth to keep R below 1% is about 1.2 GHz as seen in Fig. 4.

Oblique incidence For waves incident on the windows at an oblique angle θ, equations 1 and 2 still hold, except that now the phase change for waves going cross the window is modified to $\delta = 2\pi n d \cos\theta/\lambda_0$. The reflectivity as a function of angle is shown in Fig. 5.

While the microwaves are launched at normal incidence to the window given the relative orientation of the horns and windows, the oblique incidence effect can be significant when the waves get reflected from metallic objects, such as the vacuum chamber, or if the wavefronts themselves are curved. We will now estimate the effect due to curvature, while the chamber reflections are considered in the next section.

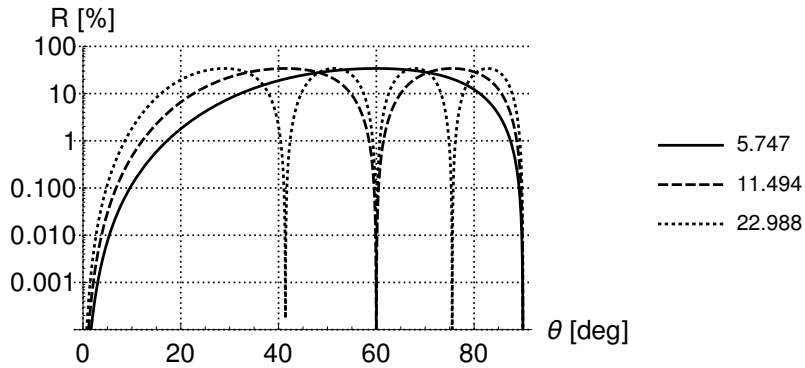

Figure 5: Residual reflectivity versus angle of incidence. The legend shows the three window thicknesses.

For a Gaussian beam of waist w_0 at a distance z away from the waist, we have the beam radius of curvature and width [5, eqn 2.21]:

$$R = z + \frac{1}{z}\left(\frac{\pi w_0^2}{\lambda}\right)^2; \quad w = w_0\sqrt{1 + \left(\frac{cz}{\pi w_0^2 f}\right)^2} \tag{3}$$

The beam intensity at a transverse position r is [5, eqn 2.25]:

$$I = I_0\left(\frac{w_0}{w}\right)^2 \exp\left(-\frac{2r^2}{w^2}\right). \tag{4}$$

The angle of incidence θ and the transverse position r are related via the radius of curvature R as $r = R\cos(\pi/2 - \theta)$. Thus we can plot the beam intensity as a function of the angle of incidence on the same plot as reflectivity vs angle (Fig 6). The beam intensity decreases with increasing distance from the beam axis (and hence, with angle of incidence). At the same time, the reflectivity increases.

The outer parts of the beam will be reflected much more than the beam centre. However, the beam intensity near the outer parts of the window is greatly reduced since most of the beam power is near the centre. At angles where we would predict a high reflectivity, the Gaussian is exponentially suppressed. Thus, the actual power reflected from the edges of the windows due to the beam curvature is negligible.

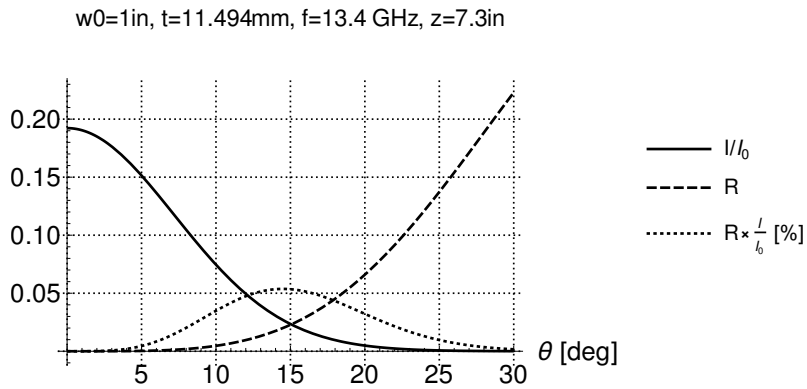

Figure 6: Residual reflectivity due to wavefront curvature.

1.2.2 Clipping

Ideally, perfectly Gaussian beams would be used with infinite windows. In practice, the vacuum chamber geometry requires the windows to be a finite size. Inevitably, a fraction of the power of the beam would get clipped at the edges of the windows. In this section, we calculate the diameters of the windows needed to lose an acceptable amount of the beam power.

According to eqn 4, the beam intensity decays exponentially as $\exp(-2r^2/w^2)$ with the transverse position r away from the beam axis. If we want the intensity at the edge of the windows to be 0.25% of the intensity at the center of the beam, then the window diameter D needs to be

$$D > 2w\sqrt{-\frac{\ln 0.0025}{2}} \approx 3.5\ w,$$

where the beam width is given in eqn 3. The vacuum chamber as built (Fig. 7) has three pairs of identical microwave windows. The two holes for 13.4 GHz microwaves have a diameters of 8.150" and are separated by 14.58", while the other two pairs (for 26.8 and 40.1 GHz) both have a 5.450" diameter and are separated by 16.13". The chamber also has a pair of 2" holes for a laser, as well as the connection for the molecular beam, perpendicular to all the laser and microwave beams.

The vacuum connection is an ISO200 F-style flange on one side, and a custom "Input Nipple" on the other (3/4" long nipple, 2.87" ID, flange 5.8" OD and 1/4" thick). The custom flange was originally intended to be used as a part of a sliding vacuum seal to allow

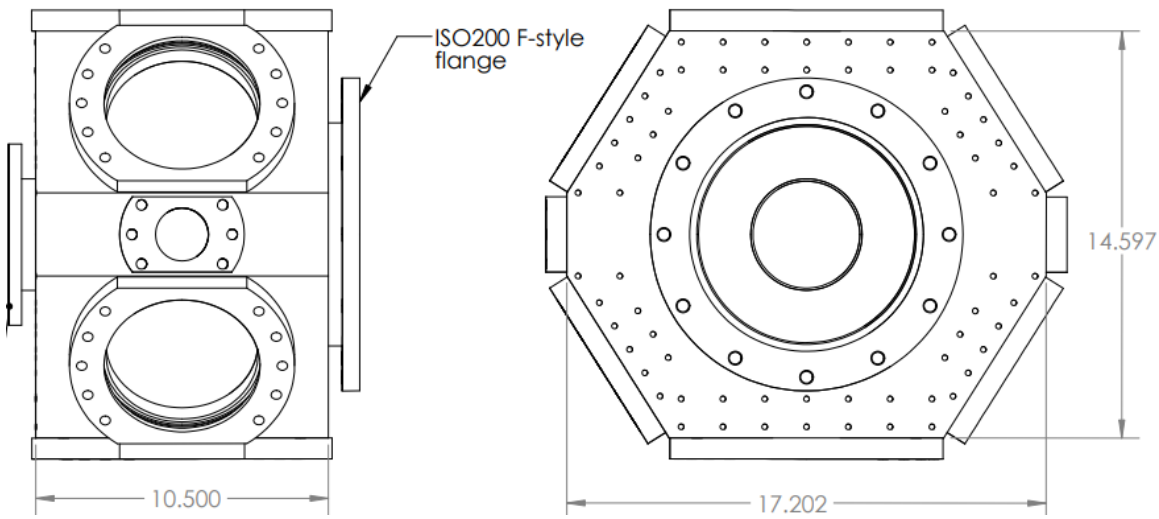

Figure 7: RC chamber

for beamline alignment while being very compact in size. The sliding seal never got designed or made, and instead we use bellows to obtain the same flexibility.

1.2.3 Beam-blocking mesh

The microwaves in the rotational cooling region interfere with other experimental regions further down the beamline and need to be blocked from entering those regions without blocking the molecular beam. This can be accomplished using a metallic mesh.

We model the mesh as two perpendicular array of parallel wires, since "the behavior of the two-dimensional grid can be calculated purely from that of its constituent one-dimensional inductive grid" [5, p. 242].

Consider infinite-length wires of radius a, permeability μ_1, and conductivity $\sigma = 1/\rho$, arranged in an array with spacing d. The reflection coefficient for radiation of frequency $\omega = 2\pi f$ and wavelength $\lambda = c/f$ at normal incidence is [6, eqns 24–25]

$$R = \frac{-1}{1 + 2Z_{\mathrm{g}}/Z_0}, \quad Z_{\mathrm{g}} = iZ_0 \frac{d}{\lambda}\left[\ln \frac{d}{2\pi a} + F(d/\lambda, 0)\right] + Z_{\mathrm{i}} d,$$

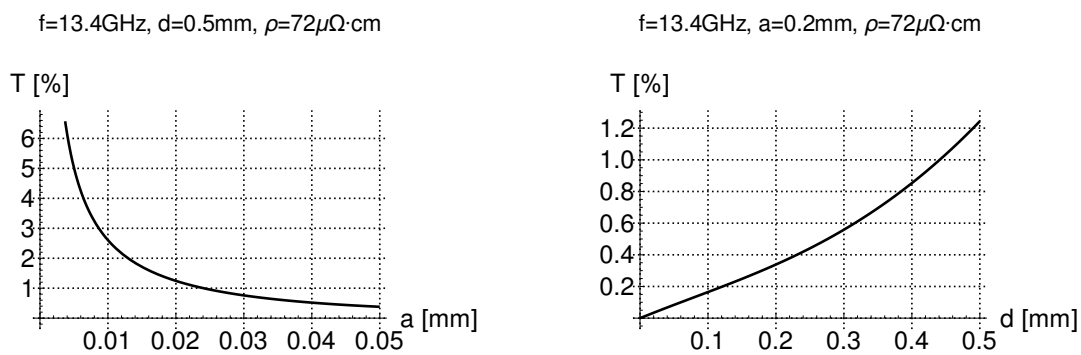

Figure 8: Transmission of 13.4 GHz microwaves at normal incidence through a wire mesh.

where $Z_0 \approx 376.7\ \Omega$ is the impedance of free space and [6, eqn 17 and 22],

$$Z_{\rm i} \approx \sqrt{\frac{\mu_1 \omega}{2\sigma}} \left(\frac{1+i}{2\pi a} \right), \quad F\left(\frac{d}{\lambda}, 0\right) = \sum_{m=1}^{\infty} \Big(\frac{1}{\sqrt{m^2 - (d/\lambda)^2}} - \frac{1}{m} \Big).$$

The transmission $T \approx 1 - |R|^2$ of 13.4 GHz microwaves through a grid is shown in Fig. 8. The resistivity of the austenitic (non-magnetic) 304 stainless steel is $\rho = 72\ \mu\Omega \cdot \text{cm}$ [7, p. 12-234], while the permeability is close to the vacuum value ($\mu_0 \approx 4\pi \cdot 10^{-7}$ H/m).

1.3 Microwave system

References

[1] Dmitry Budker, Derek Kimball, Derek F Kimball, and David P DeMille. *Atomic physics: an exploration through problems and solutions*. Oxford University Press, USA, 2004.

[2] Dean A Wilkening, Norman F Ramsey, and Daniel J Larson. Search for P and T violations in the hyperfine structure of thallium fluoride. *Physical Review A*, 29(2):425, 1984. doi:10.1103/PhysRevA.29.425.

[3] John H Moore, Christopher C Davis, and Michael A Coplan. *Building scientific apparatus*. Cambridge University Press, 2009.

[4] H Angus Macleod. *Thin-film optical filters*. CRC press, 2017.

[5] Paul F Goldsmith et al. *Quasioptical systems*. Chapman & Hall New York, NY, USA, 1998.

[6] James R Wait. Reflection at arbitrary incidence from a parallel wire grid. *Applied Scientific Research, Section B*, 4(1):393–400, 1955. doi:10.1007/BF02920016.

[7] William M Haynes. *CRC handbook of chemistry and physics*. CRC press, 97 edition, 2016.